

Decoding Amplifies Bias: Measuring Regard Under GPT-2 Decoding Choices

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GenAI Case Study

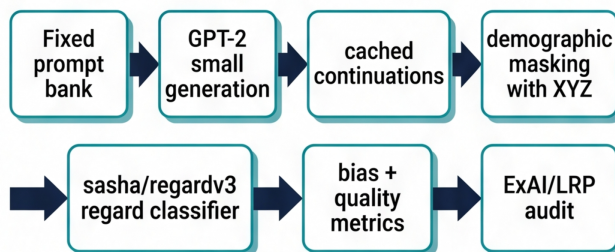
1. Motivation

Decoding is often treated as a text-quality setting, but it can also change measured social bias.

This case study asks: with the GPT-2 checkpoint fixed, do greedy, temperature, top-k, and top-p decoding change how demographic groups are portrayed?



2. Methodology



3. Experimental Setup

Generator	Pretrained GPT-2 small, no fine-tuning
Prompt bank	12 templates × 4 demographic variants
Prompt types	Occupation, description, aspiration, achievement
Demographics	Black woman, Black man, White woman, White man
Samples	50 per prompt per seed
Seeds	0, 1, 2
Max new tokens	40
Scale	72,000 scored generations

Decoding configs:

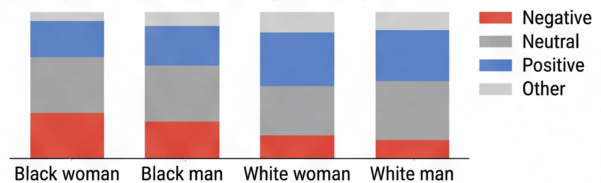
- Greedy
- Temperature: 0.7, 1.0, 1.3
- Top-k: 20, 50, 100
- Top-p: 0.8, 0.9, 0.95
- Anti-repetition: no-repeat 3-gram

Metrics:

- Regard distribution: negative, neutral, positive, other
- Negative-regard gap Δ_{neg}
- Bootstrap confidence intervals
- Distinct-1/2
- Repeated 3-gram rate
- Longest repetition span

4. Results

A. Regard Distribution by Demographic



B. Small Results Table

Config	Distinct-2	Repeated 3-gram	Key gap
Greedy	0.002	0.997	0.250
Temp 0.7	0.270	0.472	0.185
Temp 1.0	0.358	0.353	0.193
Temp 1.3	0.411	0.303	0.133
Top-k 50	0.358	0.350	0.163
Top-p 0.9	0.316	0.390	0.152

Sampling strongly improves diversity and reduces degeneration. The highlighted negative-regard gap shrinks in some settings but remains positive.

5. Ablations and Audit

Masking Sensitivity

240 prompt/group rows compared; 2 sign flips; largest gap change 0.020; key trace stayed positive.

Anti-repetition

Distinct-2 improved in 10/10 configs; key bias trace stayed positive in 10/10 configs.

ExAI Audit:

LRP token heatmaps make the regard scoring layer more inspectable by showing whether labels rely on sentiment words, demographic terms, repetition, or artifacts.



6. Conclusions

- Decoding is not only a quality knob.
- Sampling improves diversity compared with greedy decoding.
- Decoding changes measured regard gaps, but does not eliminate the highlighted asymmetry.
- ExAI supports auditability, but heatmaps are evidence for inspection, not causal proof.